

UEC867: MODERN CONTROL THEORY

L	T	P	Cr
3	0	0	3.0

Course Objective: This course provides the insight of the fundamentals of modern control theory by analysing time and frequency response of open and closed loop systems. Furthermore, the concept is extended to advanced concepts of modern control theory - centred on the system stability and state space methods. Emphasis is placed on concepts of controllability and observability of control systems.

Syllabus

Mathematical Models, Block Diagrams and Signal Flow Graphs of Systems: Introduction of mathematical models and transfer function, construction and reduction of block diagram and signal flow graphs, application of Mason's gain formula.

Time-Domain Analysis of Control Systems: Transient and steady state response, time response of first and second-order systems, steady-state errors, types of systems and error constants.

System Stability: Conditions for stability of linear systems, algebraic stability criteria, Routh criterion, root locus techniques, frequency domain analysis, correlation between frequency response and transient response, stability analysis of the control systems using various methods.

Classical Controllers: General aspects of the closed-loop control design problem, P, PD, PI and PID controllers.

State Variable Analysis: Introduction, state variable representation, conversion of transfer function model to state variable model, conversion of state variable model to transfer function model, state equations, concepts of controllability and observability,

Familiarization with standards: IEEE 3001.11 for controller application, IEEE 1547 for distributed energy resources, and the "Color Books" (e.g., IEEE 399) for system reliability.

Laboratory Work:

Course Learning Objectives (CLO)

The students will be able to:

1. Interpret closed-loop and open-loop control system representations using block diagrams and signal flow graphs.
2. Evaluate the time and frequency response of control systems and establish the relationship between them.
3. Analyze the stability of the control systems and learn various methods to judge the stability criterion.
4. Understand the fundamentals of designing of P-I-D controllers.
5. Demonstrate an understanding of state-space analysis concepts, and apply the principles of controllability and observability to both classical and digital control systems.

Text Books

1. Nagrath, I. J., and Gopal, M., *Control Systems Engineering*, New Age International Publishers, 2006, 4th ed.
2. Benjamin C. Kuo, *Automatic Control Systems*, Pearson education, 2003
3. Ogata, Katsuhiko, *Modern Control Engineering*, Prentice-Hall, (2010) 5th ed.

Reference Books

1. Warwick, Kevin, *An Introduction to Control Systems*, World Scientific Publishing Co. Pvt. Ltd, (1996) 2nd ed.
2. Levine, W. S., *Control System Fundamentals*, CRC Press, (2000) 3rd ed.
3. Mutambara, Arthur G. O., *Design and Analysis of Control Systems*, CRC Press, (1999) 2nd ed.